

# HANSIKA CHOWDARY

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## Professional Summary

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Senior Data Engineer with 4+ years of experience designing, developing, and optimizing scalable ETL/ELT pipelines, cloud data platforms, and enterprise data warehouse solutions using Snowflake, Informatica PowerCenter, Informatica Intelligent Cloud Services (IICS), Apache Airflow, and AWS. Experienced in building high-performance data integration solutions using Python, PySpark, SQL, and T-SQL, developing Airflow DAGs, automating cloud-based workflows, and processing large-scale structured and semi-structured datasets. Strong expertise in Snowflake data warehousing, AWS Glue, Amazon S3, AWS Lambda, data modeling, SQL performance tuning, and data quality, delivering secure and analytics-ready data solutions. Skilled in healthcare data integration, including HL7, FHIR, and X12 EDI, while ensuring HIPAA/HITECH compliance. Proven ability to collaborate in Agile/Scrum environments, implement CI/CD best practices, and deliver reliable, production-ready data engineering solutions.

## TECHNICAL SKILLS

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- **Programming Languages:** Python, SQL, T-SQL, PL/SQL, Java, PySpark
- **Cloud & AWS:** AWS S3, AWS Glue, AWS Lambda, Amazon RDS, AWS CloudWatch, AWS CloudTrail, Amazon EventBridge, AWS Step Functions, IAM
- **ETL & Orchestration:** Informatica PowerCenter, Informatica IICS, Informatica Data Quality (IDQ), Apache Airflow (DAGs), AWS Glue
- **Data Warehousing & Modeling:** Snowflake, SQL Server, Oracle, Star Schema, Snowflake Schema, SCD Type 1/2, Data Modeling
- **Big Data Technologies:** Apache Spark, PySpark, Apache Hadoop, Databricks, Delta Lake
- **Healthcare Data:** HL7, FHIR, X12 EDI (270/271, 834, 837), HIPAA, HITECH, PHI
- **Data Integration:** REST APIs, JSON, XML, CSV, Amazon S3, Data Pipelines
- **Visualization & BI:** Tableau, Power BI, Amazon QuickSight
- **DevOps & CI/CD:** Git, GitHub Actions, Jenkins, Docker, Terraform, Linux, Agile, Scrum
- **Monitoring & Performance:** AWS CloudWatch, Informatica Workflow Monitor, Snowflake Query History, SQL Performance Tuning, Query Optimization

## Work Experience

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### Commonwealth of Massachusetts, EOHHS

Feb 2026 - Present

#### *Senior Data Engineer, Boston, MA*

- Developed and maintained scalable ETL/ELT pipelines using Informatica PowerCenter, Informatica Intelligent Cloud Services (IICS), and Apache Airflow, ingesting healthcare data from multiple source systems into the Snowflake Cloud Data Platform.
- Developed and maintained Apache Airflow DAGs to orchestrate end-to-end ETL workflows, automating data ingestion, transformation, and loading between AWS S3, Informatica IICS, and Snowflake.
- Designed and optimized complex Snowflake SQL, T-SQL, stored procedures, views, CTEs, and window functions to improve query performance and support enterprise reporting and analytics.
- Built and maintained Snowflake data models, schemas, secure views, fact tables, and dimension tables, implementing Star Schema and Slowly Changing Dimensions (SCD Type 1/2) for enterprise data warehousing.
- Developed automated data ingestion pipelines using AWS S3, Python, and PySpark, processing structured and semi-structured data including JSON, XML, and CSV for downstream analytics.
- Implemented comprehensive data quality, validation, reconciliation, and exception handling using Informatica Data Quality (IDQ), SQL, and Python, ensuring accurate, consistent, and trusted healthcare datasets.
- Integrated healthcare data from Oracle, SQL Server, REST APIs, HL7, FHIR, and X12 EDI sources, enabling secure and compliant enterprise data exchange.
- Monitored, troubleshooted, and optimized production ETL workflows using Apache Airflow, Snowflake Query History,

- AWS CloudWatch, and Informatica Workflow Monitor, ensuring high availability and reliable pipeline execution.
- Automated workflow scheduling and cloud orchestration using AWS Lambda, Amazon EventBridge, Git, and CI/CD pipelines, improving deployment efficiency and operational reliability.
  - Collaborated with data analysts, QA teams, application developers, and business stakeholders in Agile/Scrum environments to gather requirements, implement data solutions, and support SDLC release activities.
  - Ensured compliance with HIPAA and HITECH standards while maintaining source-to-target mappings, metadata, technical documentation, role-based access controls, and enterprise data governance within Snowflake.

## **Tetra Tech**

**Jan 2025 – Jan 2026**

*Data Engineer, Los Angeles, CA*

- Developed and maintained scalable ETL pipelines using AWS Glue, Informatica PowerCenter, SQL Server, and Amazon S3, processing high-volume enterprise data from multiple source systems.
- Designed and optimized complex T-SQL queries, stored procedures, views, CTEs, window functions, and indexing strategies to improve query performance and support enterprise reporting.
- Processed and integrated structured and semi-structured data from EDI, JSON, XML, CSV, and flat files, ensuring accurate, reliable, and scalable data ingestion into enterprise data warehouses.
- Built cloud-native data ingestion and transformation pipelines using AWS Glue, PySpark, Python, AWS Lambda, and Amazon S3, enabling scalable processing of large-volume datasets.
- Implemented data quality, reconciliation, and validation frameworks using Python, SQL, and Informatica Data Quality (IDQ), identifying duplicate records, schema inconsistencies, and referential integrity issues.
- Designed dimensional data models using Star Schema, Fact and Dimension tables, surrogate keys, and Slowly Changing Dimensions (SCD Type 1/2) to support enterprise analytics and business intelligence.
- Integrated data from Oracle, SQL Server, Amazon RDS, REST APIs, Amazon S3, and flat files, enabling seamless data movement across on-premises and AWS cloud environments.
- Optimized ETL workflows and database performance through query tuning, execution plan analysis, indexing, partitioning, caching, and parallel processing, significantly improving pipeline efficiency.
- Automated workflow orchestration and deployments using AWS Step Functions, Amazon EventBridge, Git, Jenkins, and CI/CD pipelines, ensuring reliable and repeatable data integration processes.
- Developed curated datasets and data marts supporting Amazon QuickSight, Tableau, and Power BI dashboards, delivering actionable insights and enterprise reporting.
- Monitored production ETL workflows using Amazon CloudWatch, AWS CloudTrail, and Informatica Workflow Monitor, performed root cause analysis, maintained technical documentation, and implemented data governance best practices to ensure reliable pipeline execution.

## **University of Maryland, Baltimore County**

**Jan 2024 – Dec 2024**

*Graduate Research Assistant, Baltimore, MD*

- Developed scalable data processing pipelines using Python, ingesting, transforming, and structuring large-scale datasets for enterprise analytical and operational workloads.
- Designed and optimized ETL pipelines using Databricks and SQL, enabling efficient data ingestion, transformation, and loading into enterprise cloud data platforms.
- Leveraged Apache Spark (PySpark) for distributed data processing, enabling scalable handling of large-scale big data workloads and improving processing performance.
- Built and optimized complex SQL queries and data transformation logic, improving query efficiency, reducing execution time, and supporting high-performance analytical workloads.
- Processed structured and semi-structured data using Apache Hadoop and distributed computing frameworks to support enterprise-scale big data applications and analytics.
- Integrated data from multiple heterogeneous sources into Delta Lake using Databricks, ensuring reliable, scalable, and ACID-compliant transactional data processing.
- Collaborated with analytics teams to deliver curated datasets supporting Tableau and Power BI dashboards, enabling actionable business intelligence, reporting, and data-driven decision-making.

## **Cognizant Technology Solutions**

**Mar 2021 - Dec 2022**

*ETL Developer, Hyderabad, India*

- Developed ETL pipelines using Informatica PowerCenter, extracting and transforming data from Oracle, SQL Server, and flat files into enterprise data warehouses.
- Designed and optimized ETL workflows using AWS Glue, enabling efficient data ingestion, transformation, and loading into Amazon S3 and Snowflake.
- Built and optimized complex SQL and PL/SQL queries, stored procedures, and functions, improving ETL performance and data consistency.
- Leveraged PySpark for distributed data processing, enabling scalable transformation of large-volume datasets.
- Implemented data validation, reconciliation, and quality checks using Python, ensuring schema consistency and data integrity across pipelines.
- Designed dimensional data models in Snowflake, improving query performance and supporting enterprise reporting requirements.
- Integrated data from relational databases, cloud storage, APIs, and external sources using Azure Data Factory, enabling seamless enterprise data movement.
- Collaborated with cross-functional teams to deliver data solutions supporting Tableau and Power BI dashboards for business reporting.
- Performed ETL testing, debugging, and production support using AWS CloudWatch and SQL validation techniques, ensuring reliable and high-performing data pipelines.

## **Education**

**University of Maryland, Baltimore, MD**

**Jan 2023 – Dec 2024**

*Master of Science, Data Science*

**Matrusri Engineering College, Hyderabad, India**

**June 2017 – May 2021**

*Bachelor of Science, Computer Science and Engineering*